Bilkent University

Department of Computer Engineering

Senior Design Project

Analysis Report

Bil-Events : A mobile app to organize events in Bilkent University

Team Members

Osman Can Yıldız -21302616

Kasım Sarp Ataş - 21502006

Can Savcı - 21300803

Supervisor: Halil Altay Güvenir

Jury Members: Fazlı Can and Özgür Ulusoy

Innovation Expert: Barış Misman

Analysis Report
March 23, 2020

This report is submitted to the Department of Computer Engineering of Bilkent University in partial fulfillment of the requirements of the Senior Design Project course CS491/492.

Table of Contents

1. Introduction

2. Current System

3. Proposed System
    3.1. Overview
    3.2. Functional Requirements
    3.3. Nonfunctional Requirements
        3.3.1. Usability
        3.3.2. Performance
        3.3.3. Reliability
        3.3.4. Maintainability
    3.4 Pseudo Requirements
    3.5 System Models
        3.5.1 Scenarios
        3.5.2 Use Case Model
            3.5.2.1 Login Use Case
            3.5.2.2 Change Password Use Case
            3.5.2.3 See Clubs Segment Use Case
            3.5.2.4 Recommendation Segment Use Case
            3.5.2.5 Logout Use Case
            3.5.2.6 SeeEvents Use case
        3.5.3 Object and Class Model
        3.5.4 Dynamic Models
            3.5.4.1 Application Flow Activity Diagram
            3.5.4.2 Sequence Diagram
            3.5.4.3 Sequence Scenario: The User Tries To Login
            3.5.4.4 Sequence Method: The User Searches for an Event and Finds Information about the Event
            3.5.4.5 Sequence Method: The User Attends to an Event and Recommendation Page Update
            3.5.4.6 Sequence Method: The User Searches for an Event among the Recommended Events and Chooses an Event
        3.5.5. User Interface - Navigational Paths and Screen Mock-ups
            3.5.5.1. LOGO
            3.5.5.2. Login Page
            3.5.5.3. Homepage
            3.5.5.4. Club List
            3.5.5.5. Upcoming Events
            3.5.5.6. Selected Events
            3.5.5.7. Recommended Events

4. Other Analysis Elements
    4.1. Consideration of Various Factors
        4.1.1 Social Factors
        4.1.2. Environmental Factors
        4.1.3. Economic Factors

4.2. Risks and Alternatives
4.2.1. Failure to Provide Sufficient Accuracy Rate
4.2.2. Scheduling Risks
4.2.3. Leak of Private Data
4.2.4 Potential Cheating from Club Members
4.3. Project Plan
4.4. Ensuring Proper Team-Work
4.5. Ethics and Professional Responsibilities
4.6. New Knowledge and Learning Strategies

5. References

1. Introduction

As a better organization is key for events, we are trying to do an organizer application for Bilkenters. A mobile app called Bil-Events collects and organizes the activities better than the current system which are posters on the boards, BAIS (Bilkent University Administrative Information System), social media and BSO website in Bilkent University. This app will be done with the purpose of easy access to events in Bilkent, making activities more organized and notifying Bilkenters in a better way. In addition, getting more benefits from clubs,increasing the competition between clubs, getting more participants to clubs, increasing the interactions and collaborations among students and lastly pushing club leaders to be more active is the expected positive effect of Bil-Events. Another expectation is to reduce paper consumption caused by posters. For each event, clubs stick posters everywhere, which is also not equally distributed for every poster board, so poster use has a great deal of paper consumption regularly. In the tech age, papers are less accessible for people instead of apps in smartphones. Therefore, our aim is both to decrease paper consumption and organize while collecting all event notifications in one place. In addition to this, students want to improve themselves both physically and mentally; therefore, students prefer to go to events to gain the skills, to extend their networks and have fun. Club members want to reach out to the newcomers so that the sustainability of the clubs does not become an issue for the members of the clubs. Therefore, our app will provide the students to help spending their time more efficiently such as finding the location of the event quickly and effectively such as giving them the opportunity to choose their activities based on their interests and helping to take the responsibilities to change the ambiance of the Bilkent campuses.

2. Current System

Current system is too separate from each other. For instance, the events on the poster may not be notified on the social media accounts of the clubs or not be notified with BAIS mails.

Posters

Posters is a kind of notification way that should be changed for its unpractical services such as:

- Unnecessary paper consumption
- Not equally distributed on the boards in different building, event different floor and halls.
- Waste of money for clubs
- Unnecessary effort for trying to stick poster to boards in different places

BAIS (Bilkent University Administrative Information System)

BAIS is a kind of notification way that should be changed for its unpractical services such as:

- Everything about University and student come as a BAIS mail to the students' email inbox that makes the inbox to be messy and hard to remember the events and their informations.
- If students want mails from Bilkent clubs, they have to accept mail terms on the SRS system for a specific club. Then the mail inbox become more messy than ever since all club notifications start to come as an email or multiple emails.

Social Media

Social media is a kind of notification way that should be changed for its unpractical services such as:

- Clubs' social media accounts can be found and followed but this problem is how students can know the existence of clubs.
- When events are cancelled, the notification on the social media may not be reached or notified before the event.

3. Proposed System

Below we will explain our project in more detail.

3.1. Overview

Bil-Events is an app to offer essential services to club members and participants. In the participants' standpoint, Bil-Events helps them to find the suitable event for their interests, gives very detailed information about the chosen event so that the student can benefit from the opportunities of the specific events such as being on time, networking, preparing himself/herself for the event etc. A student club member can prepare a very detailed and aesthetically pleasing page about the event so that the event can be very appealing to the students who want to improve themselves. Therefore, Bil-Events is going to collect the data of events and participating students' information so that club members can see the success / failure rate of their advertisement. In order to improve the quality of the services, Bil-Events is going to collect the data of previously gone events of the specific student in order to find the student's preferences on upcoming events. Therefore, the embedded recommender system on Bil-Events app can help a student to recommend a suitable event according to the student's interest. In addition to this, notifications of the upcoming events will be given to the students. In the club members' standpoint, if a student attends the events hosted by the club, they will be notified so that the club members can meet with the passionate student. In addition to this, the students can like/dislike the previously happened events and make comments about them.

3.2. Functional Requirements

- Users will create accounts with their student ID's.
- The users which are not in the Student Database Management System of the Bilkent University cannot create an account.
- Users will be asked to specify their areas of interests (e.g. art, music, etc) that will be used for suggesting events in the recommendation process.
- The users can search for the events and seminars in Bilkent.
- The user can select an event to see details such as description, location, starting time etc.
- Bil-Event will list events in a way that specific events which are recommended for the user will be on top.
- The user can choose to attend an event.
- The user can choose to create an event but to create an event, the user must be the official club leader.
- After the event is posted on the app, everyone can access it immediately.
- After attending an event, the user can rate the event and write reviews about his/her experiences.
- The user can rate attended events on multiple categories
- The club members can reach the information of the most passionate student who attended their events the most.

3.3. Nonfunctional Requirements

3.3.1. Usability

The application will be targeted at students and club leaders so the application's user interface will be simple and straightforward. Everyone with basic mobile knowledge should be able to use the app and the average user should become acquainted with all of the functionalities in less than 5 minutes. To this end, tutorials will be included in the app.

Since internet connection may not be accessible as it is possible, Bil-Events will work offline for the saved event' notification. For a more accurate diagnosis the users can refresh the events' information when they connect to the internet.

3.3.2. Performance

The response time for the finding events should be no more than 10 seconds if the internet connection is sufficient enough. The load time of the application should be no more than 5 seconds but it can change according to Android version and hardware capacity. The

application should take no more than 2 second to switch between panels in order to offer a smooth user experience for our users.

3.3.3. Reliability

Our application will give importance to reliable information about the students in the university. The students which are from other universities or the students with the fake accounts cannot sign in to the Bil-Events. Every student should have only one account on the application and it should be the verified account by the BCC.

3.3.4. Maintainability

The system will be developed in a way to be flexible to possible changes in the future. The code will be well documented in order to understand and fix issues that may arise faster.

3.4 Pseudo Requirements

- Java programming language will be used to develop the majority of the project.
- Android Studio will be used to build the application.
- The program will be delivered as an .apk executable.
- MySQL will be used to manage all database functionalities.
- We will use Apache PredictionIO Recommender System to embed it to our Android application.
- Git platform will be used to keep track of the application's development.
- Bil-Events will be available on Google Play Store for free.
- The application will not require internet connection to saved events,but for a more accurate result internet connection is advised if the event's information is changed.
- The application will require internet connection to access and search current events.

3.5 System Models

3.5.1 Scenarios

This section explains how the app will function during different user interactions. Importance of this section is demonstrating the events that triggers after user did some intreactioın with the application

Scenario 1

Scenario Name: Login

Participating actor instances: Ahmet: Bilkent Student

Main flow of events:

1. Ahmet has heard of Bil-Events and has never used it.
2. Ahmet downloads the application from Google Play Store and opens it.
3. Ahmet enters his username as his school id "21000000 ".
4. Ahmet enters his password given by the school.
5. Ahmet taps the "Login" button.
6. Bil-Events checks if the username is correct.
7. Bil-Events checks if the password is correct
8. Bil-Events confirms that the information Ahmet inserted is correct
9. Bil-Events redirects user Ahmet to his Home page.

Scenario 2

Scenario Name: Unsuccessful Login

Participating actor instances: Ahmet: Bilkent Student

Main flow of events:

1. Ahmet has heard of Bil-Events and has never used it.
2. Ahmet downloaded the application from Google Play Store and opened it.
3. Ahmet enters his username as his school id "21000000 ".
4. Ahmet enters his password given by the school.
5. Ahmet taps the "Login" button.
6. Bil-Events checks if the username is correct.
7. Bil-Events checks if the password is correct
8. Bil-Events confirms that the information Ahmet inserted is incorrect
9. Bil-Events pop an unsuccessful login window.

Scenario 3

Scenario Name: Change Password

Participating actor instances: Ahmet: Bilkent Student

Main flow of events:

1. After Ahmet logged into the Bil-Events, he taps Setting button
2. Ahmet saw his account information.
3. Ahmet decided to change his password.
4. Ahmet clicks the "Change Password" button.
5. Ahmet changes his password.

Scenario 4

Scenario Name: EventSelect

Participating actor instances: Ahmet: Bilkent Student

Main flow of events:

1. Ahmet is on the HomePage of the application.
2. Ahmet decides to find an upcoming event happening on campus and opens the club list.
3. Bil-Events provides Ahmet information about different Clubs.
4. User Ahmet selects a Club he is interested in.
5. Bil-Events provides Ahmet information about the Club he chose and that Club's upcoming events.
6. Ahmet decided to go to a specific event.
7. Events are added automatically into events lists of Ahmet.
8. Bil-Events saves this information for Ahmet and notifies him 24 hour ahead before the event is about to start.
9. Ahmet gets notification and prepares to go to the event.
10. Event is deleted from the saved list of Ahmet after the deadline has passed. It will also be deleted from the list of events.

Scenario 5

Scenario Name: RecommendationSystem

Participating actor instances: Ahmet: Bilkent Student

Main flow of events:

1. Ahmet is on the HomePage of the application.
2. Ahmet decides to find an upcoming event happening on campus and opens the club list.
3. Ahmet selects a Recommended part.
4. Bil-Events provides Ahmet information about different Club's events based on Ahmet's previous attended events.
5. Ahmet decided to go to a specific event.
6. Events are added automatically into events lists of Ahmet.
7. Bil-Events saves this information for Ahmet and notifies him at a 24 hour ahead before the event is about to start.
8. Ahmet gets notification and prepares to go to the event.
9. Event is deleted from the saved list of Ahmet after the deadline has passed. It will also be deleted from the list of events.

Scenario 6

Scenario Name: UserChange

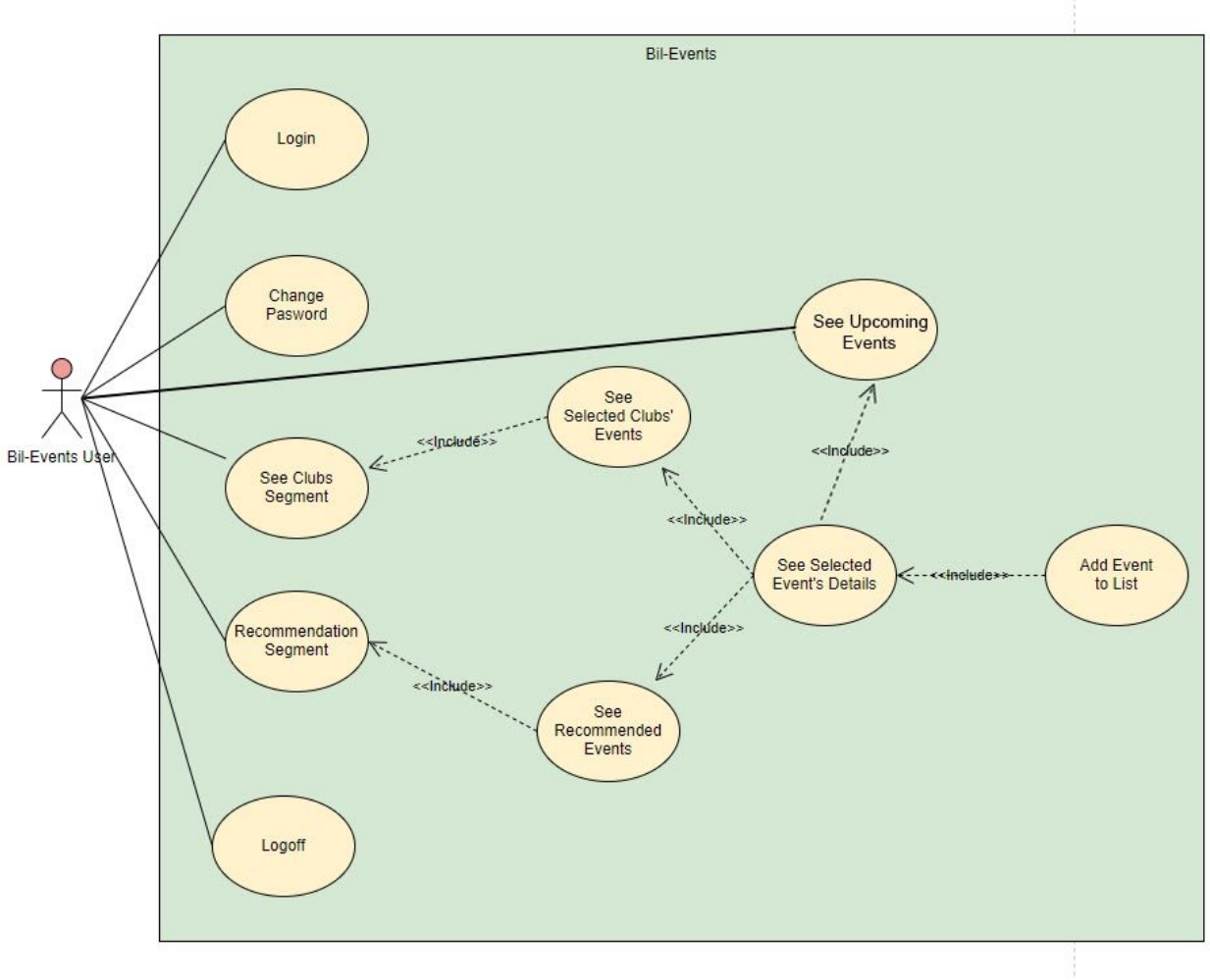
Participating actor instances: Ahmet and İdil: Bilkent Students

Main flow of events:

1. Ahmet is on the HomePage of the application.
2. While Ahmet is on the Application, İdil wants to check events on Ahmet's phone in her own account
3. Ahmet decided to logoff.
4. İdil enters the application with her own account.
5. İdil enters the app through the Scenario 1

3.5.2 Use Case Model

The following section will introduce the Use Case Diagram of our system. The reason we used the use case methodology to mainly understand our own system. How it should have work and how cases should be. This section also explains the user and system interactions. "Login" is required for all other cases except "Change Password".



3.5.2.1 Login Use Case

Use Case Name: Login

Actor: Bil-Events User

Entry Condition:

- User is on the Login Screen.

Exit Condition:

- User is on the HomePage screen.
- Authentication Failed and user is on Login screen.

Main flow of events:

1. User enters his Username.
2. User enters his password.
3. Bil-Events checks if the email and password are correct.
4. Bil-Events confirms the credentials of the user.
5. Bil-Events redirect users to HomePage.

Alternative flow of events:

1. User enters the wrong username and/or password.
    a. Bil-Events detects that the credentials are incorrect.
    b. Bil-Events redirects user to Login page
    c. Bil-Events notifies users about the wrong entries.

3.5.2.2 Change Password Use Case

Use Case Name: Change Password

Actor: Bil-Events User

Entry Condition:

- User is on Login Screen and taps the “Change Password” button.

Exit Condition:

- User is finished changing his/her password.
- User is decided not to change his/her password and press the back button.

Main flow of events:

1. User taps the “Change Password” button.
2. User enters his new password.
3. User confirms his new password.

4. Bil-Events confirms the changes of the user password and updates the database.
5. Bil-Events redirect users to HomePage.

Alternative flow of events:

1. User enters wrong password to the confirmation segment..
    a. Bil-Events detects that the both passwords are not matching.
    b. Bil-Events redirects users to Change Password page.
    c. Bil-Events notifies users about the wrong entries.

3.5.2.3 See Clubs Segment Use Case

Use Case Name: See Clubs Segment

Actor: Bil-Events User

Entry Condition:

- User is on the Homepage of the Application.

Exit Condition:

- User is on the “See Clubs Segment” Page.

Main flow of events:

1. User login to the Bil-Events
2. User taps to “See Clubs” button
3. Bil-Events redirect users to the “See Clubs” page.

Alternative flow of events:

1. User decided to exit at Home Page
    a. Bil-Events terminates.
2. User decided to tapped somewhere else
    a. Bil-Events redirect users to the desired page.

3.5.2.4 Recommendation Segment Use Case

Use Case Name: Recommendation Segment

Actor: Bil-Events User

Entry Condition:

- User is on the Homepage of the Application.

Exit Condition:

- User is on the “Recommendation Segment” Page.

Main flow of events:

1. User login to the Bil-Events
2. User taps to “Recommendation” button
3. Bil-Events redirect users to the “Recommendation” page.

Alternative flow of events:

1. User decided to exit at Home Page
    a. Bil-Events terminates.
2. User decided to tapped somewhere else
    a. Bil-Events redirect users to the desired page.

3.5.2.5 Logout Use Case

Use Case Name: Logout

Actor: Bil-Events User

Entry Condition:

- User is on the Homepage of the Application.

Exit Condition:

- User taps LogOut button

Main flow of events:

1. User taps the “Logout” button.
2. User confirms that request
3. Bil-Events redirect users to the “Login” page.

3.5.2.6 SeeEvents Use case

Use Case Name: See Club Events

Actor: Bil-Events User

Entry Condition:

- User is on the Clubs Segment of the Application.

● User is on the Recommended Events Segment of the Application.

Exit Condition:

● User taps Back button

● User selects an Event. Now User is on Event Details Segment

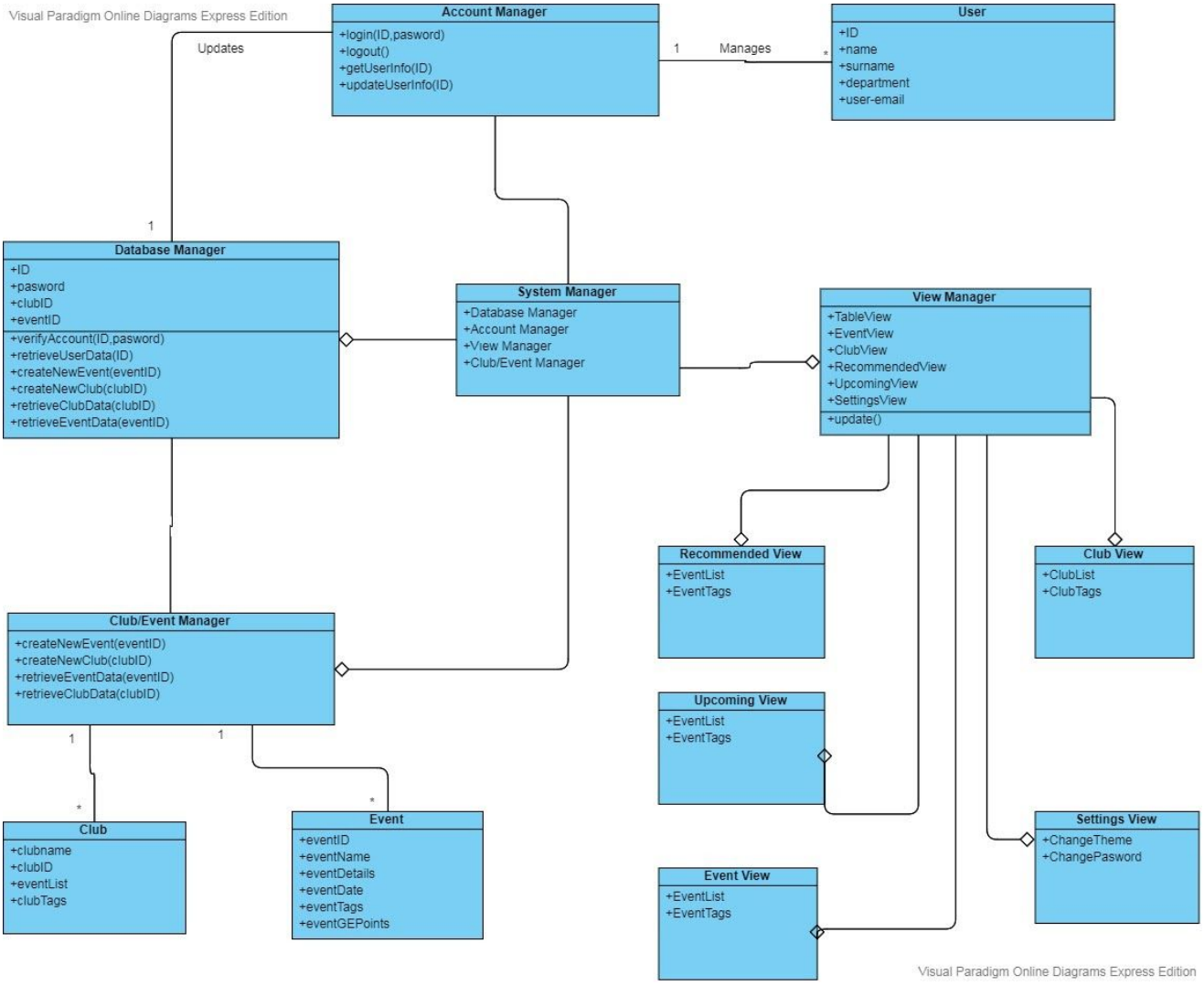
Main flow of events:

1. User taps the desired Event at Club's Event Segment.

Alternative flow of events:

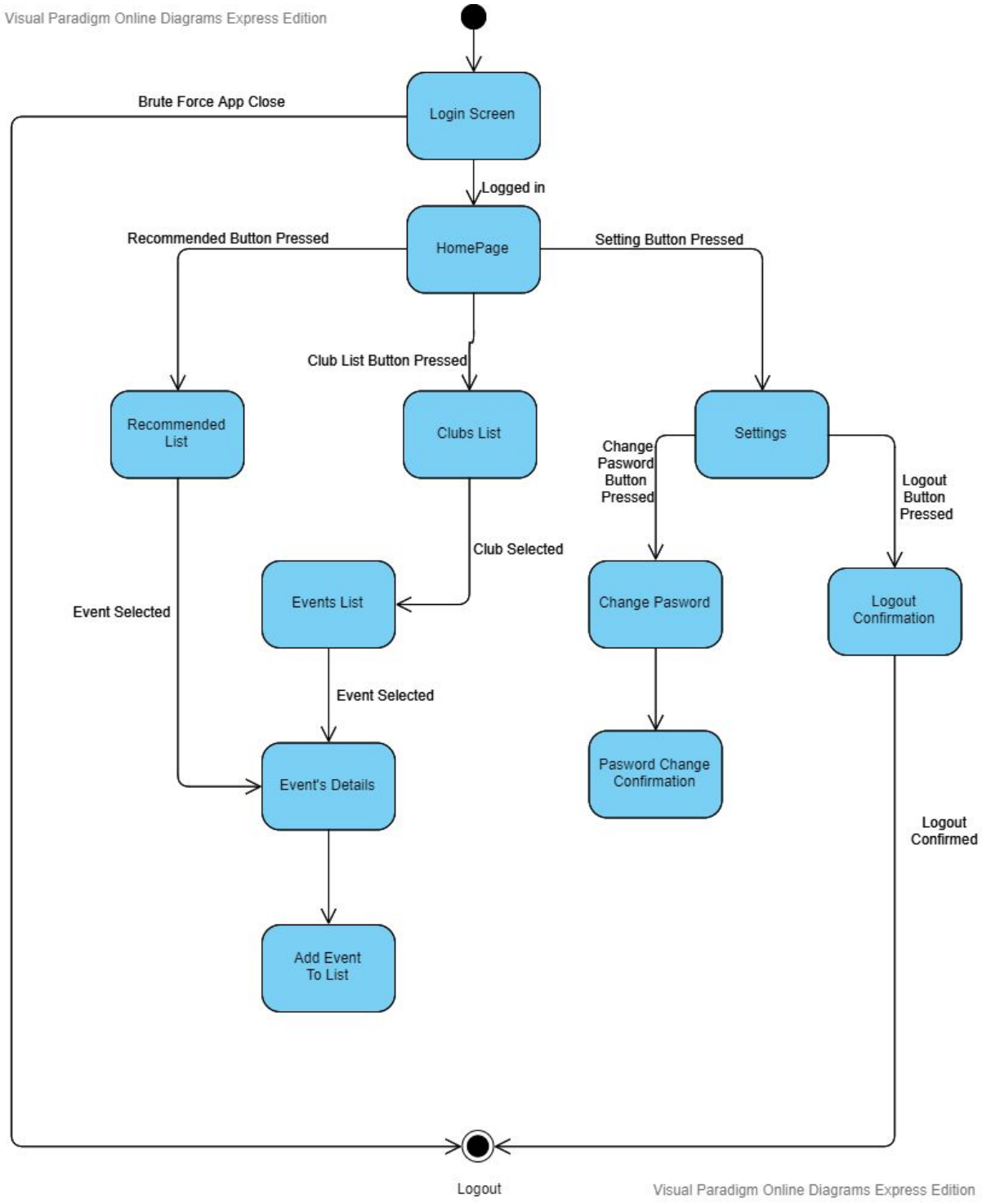
1. User decided to press Back button
    a. Bil-Events redirect users to the previous page.
2. User decided to Exit
    a. Bil-Events terminates.

3.5.3 Object and Class Model



3.5.4 Dynamic Models

3.5.4.1 Application Flow Activity Diagram



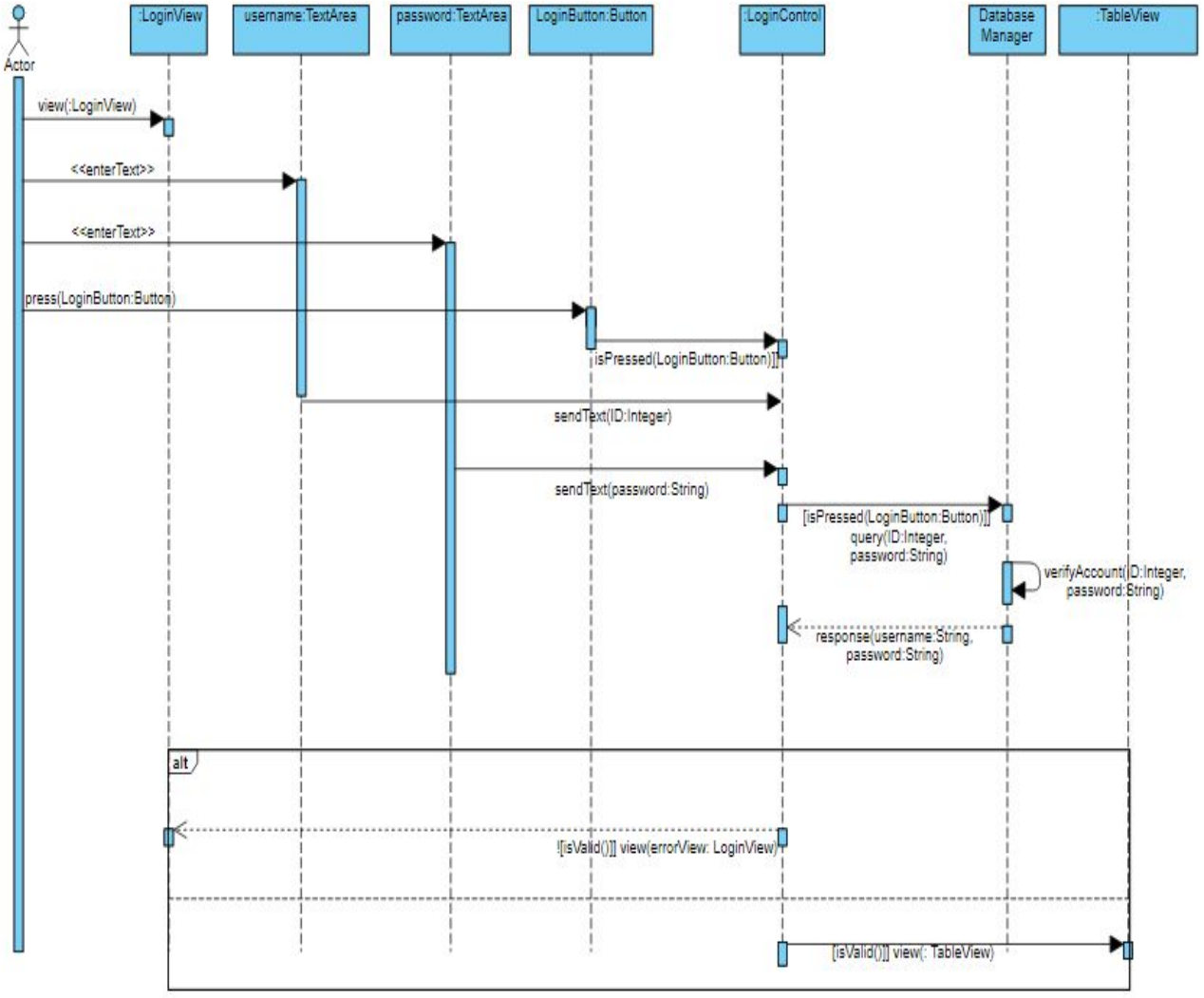
Bil-Event Application can be closed at any state by brute force. After closing the app it will automatically logout the account.

3.5.4.2 Sequence Diagram

The Sequence Diagrams for 4 possible scenarios are drawn as follows.

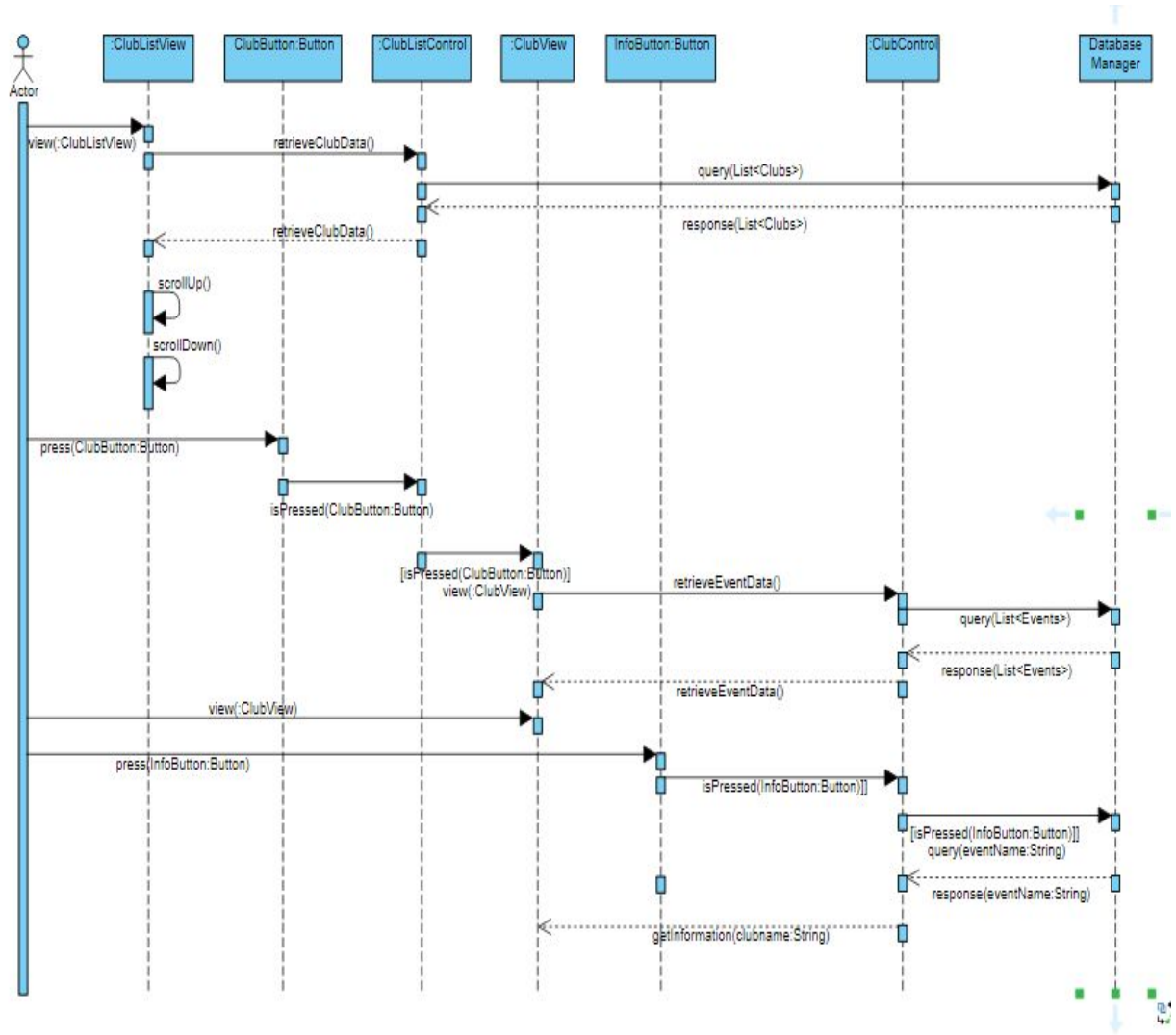
3.5.4.3 Sequence Scenario: The User Tries To Login

First, the user opens the login page. The user sees the username and password textboxes, enters its username and password. Then, the user presses the login button. The strings username and the password is sent to LoginControl. LoginControl connects with the Database Manager and checks whether the username and the password are valid or not. If they are not valid, LoginControl generates an error. If they are valid, the user can see the homepage.



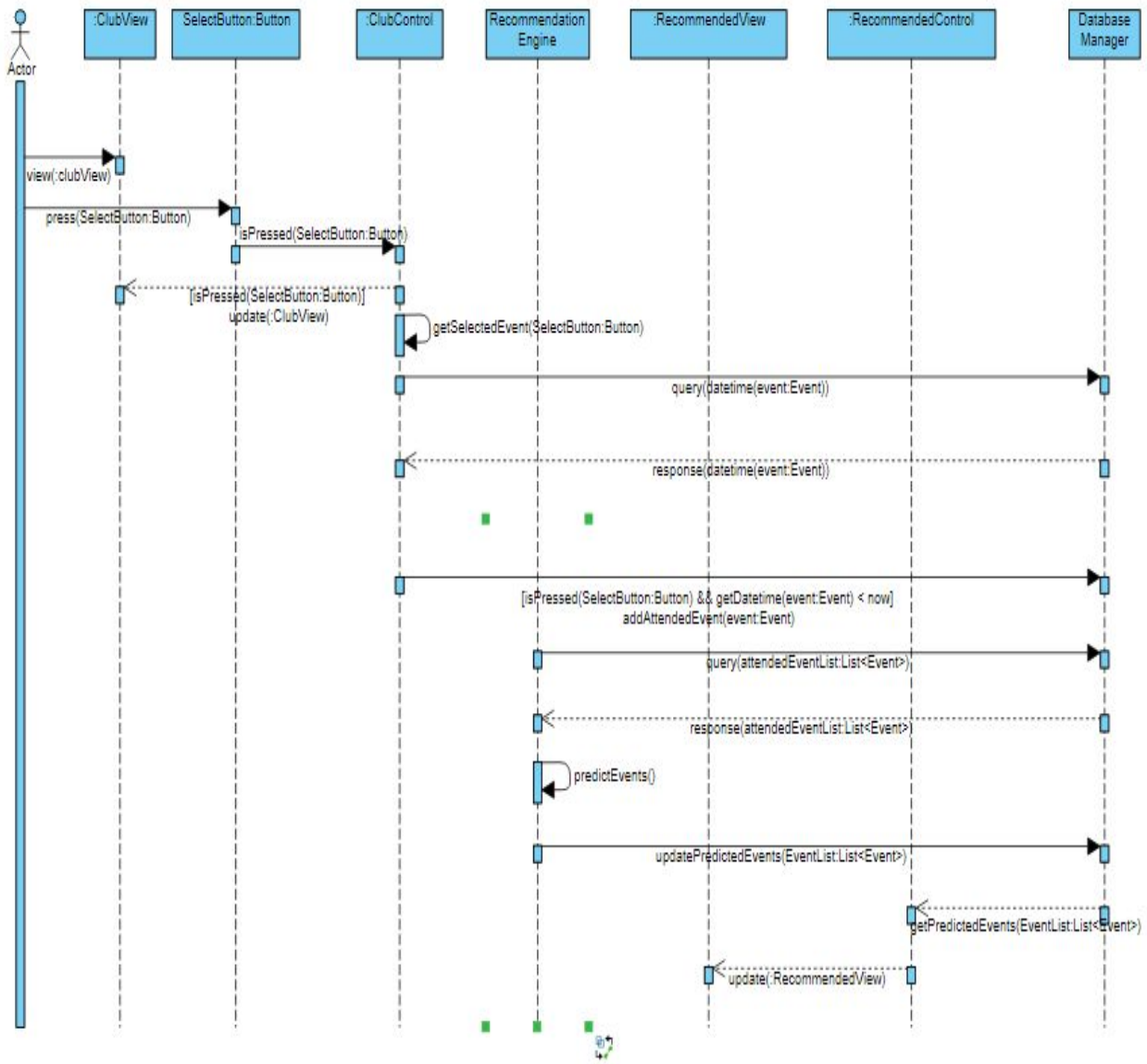
3.5.4.4 Sequence Method: The User Searches for an Event and Finds Information about the Event

The user enters the Club List Page. ClubListView sends the information to ClubListControl to view the listed clubs. ClubListControl interacts with the Database Manager to get the all clubs listed and sends them to the ClubListView page. The user can scroll up or down to find the clubs that it is looking for and when it finds a club, the user presses the button. The information of click is sent to the ClubListControl. ClubListControl interacts with the Control Manager and the user is directed to the ClubView page. The ClubView object interacts with the ClubControl to list all of the events. The ClubControl object interacts with the Database Manager to get all of the events related to the specific club and updates the ClubView object. When the user presses the Info button, ClubControl object gets the pressing event and interacts with the Database Manager to get the information about the specific event. The ClubControl object gets the information and updates the ClubView object. The user sees the information of the specific event by clicking the Info button.



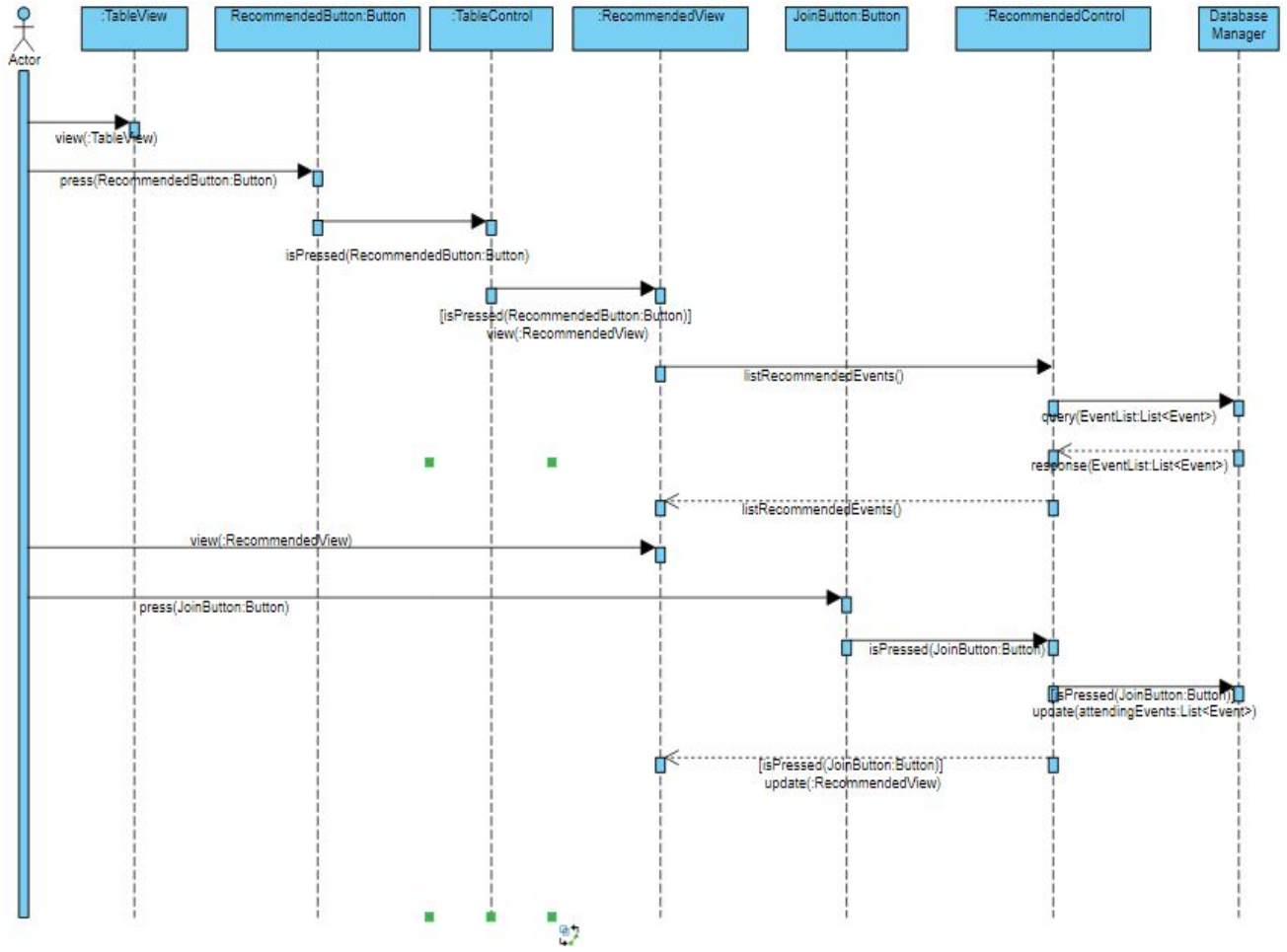
3.5.4.5 Sequence Method: The User Attends to an Event and Recommendation Page Update

The user views the ClubView page and sees lots of events in the ClubView page. The user chooses the event from one of them. The ClubControl object takes the event of pressing the SelectButton and updates the ClubView page first. Then, the ClubControl object determines the selected event. The ClubControl object interacts with the Database Manager and gets the datetime of the chosen event after the event is started. If the user clicks the SelectButton of the specific event and the event is already started, the ClubControl object interacts with the Database Manager and adds the event to the attended events list. After that, the Recommendation Engine interacts with the Database Manager and gets the updated list of attended events. The Recommendation Engine predicts the events that the user can potentially choose to attend. After obtaining the predicted event list, Recommendation Engine interacts with the Database Manager to update the table of predicted events. The RecommendedControl object interacts with the Database Manager to get the predicted events and updates the RecommendedView object. After that, the user can see the updated recommended event list.



3.5.4.6 Sequence Method: The User Searches for an Event among the Recommended Events and Chooses an Event

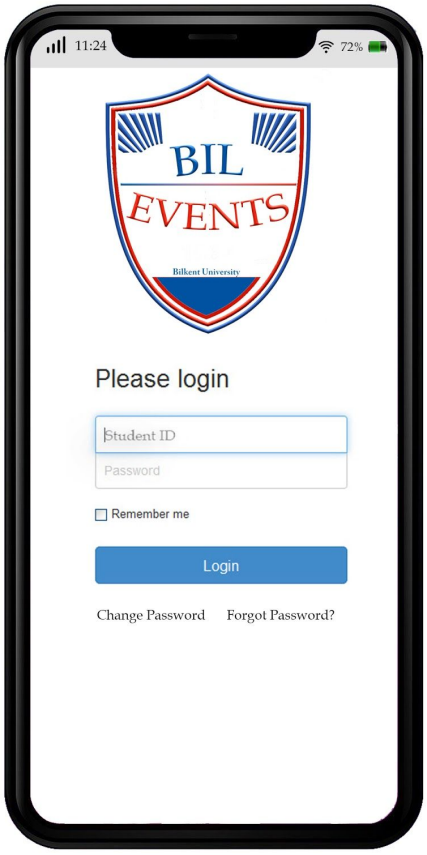
The user opens the home page and presses the Recommended Events option in the home page. TableControl object takes the event of pressing RecommendedButton object and opens the RecommendedView page. RecommendedView object interacts with the RecommendedControl object and requests the list of recommended events stored in the database. The RecommendedControl object interacts with the Database Manager and gets the list of recommended events. Then, the RecommendedControl object updates the RecommendedView object. The user opens the Recommended Events page and sees the recommended events. After that, the user chooses an event to attend and presses JoinButton. The RecommendedControl object takes the event of pressing the JoinButton. The RecommendedControl object interacts with the Database Manager to update the list of attending events, the object updates the RecommendedView object as well. The RecommendedControl object deletes the selected event.



3.5.5. User Interface - Navigational Paths and Screen Mock-ups

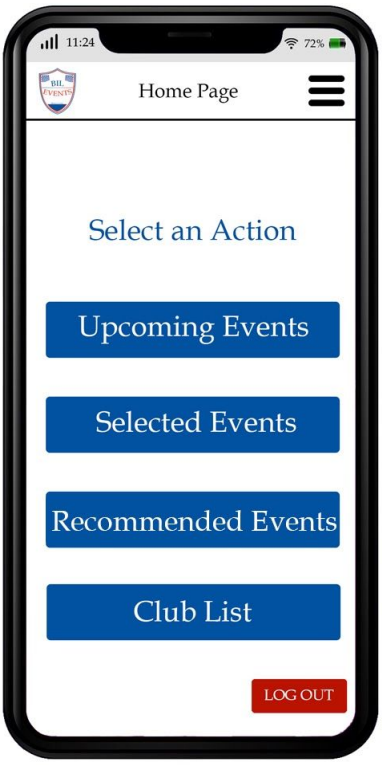
3.5.5.1. LOGO

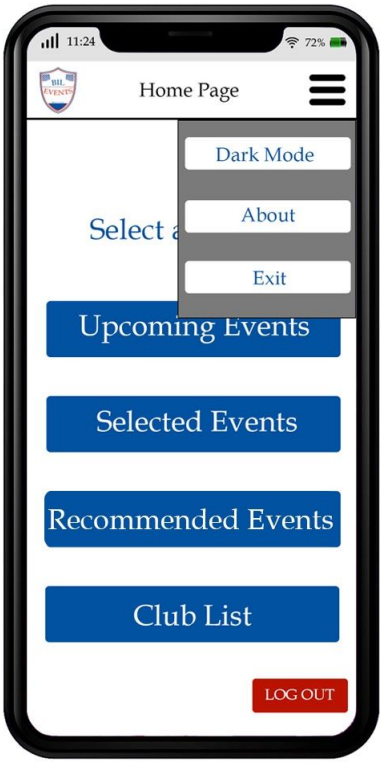
3.5.5.2. Login Page



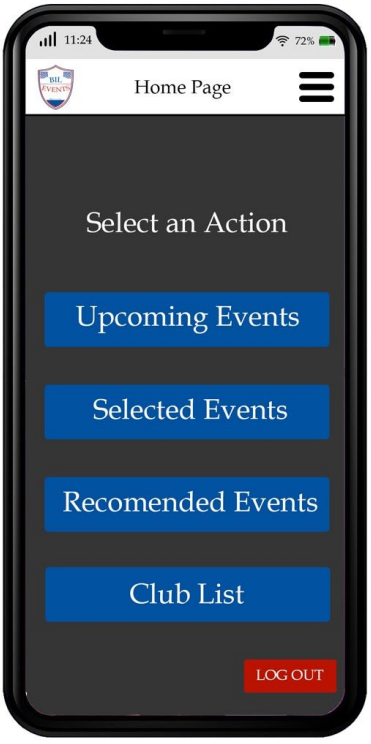
3.5.5.3. Homepage

Original Home Page



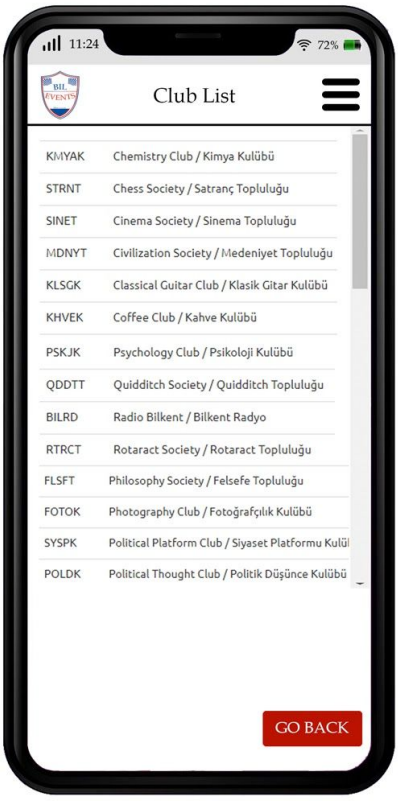


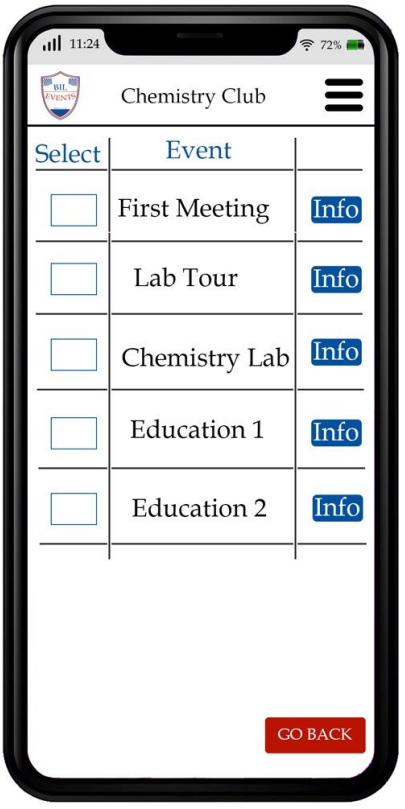
Dark Mode Home Page

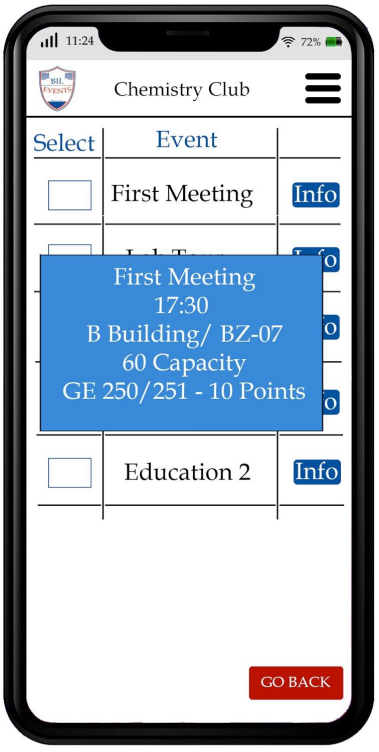


About Page In Home Page

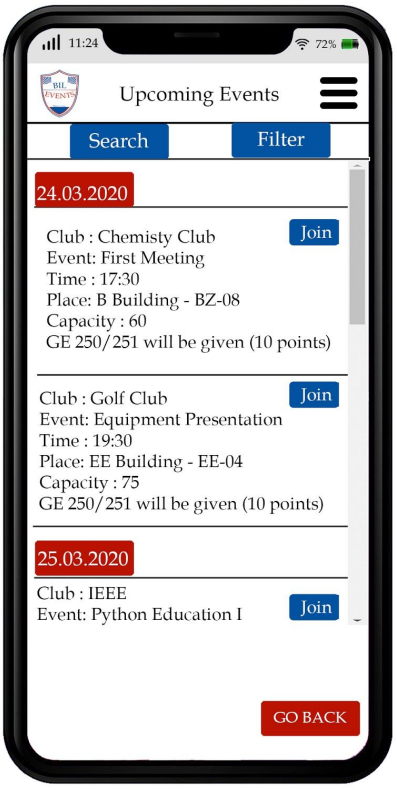
3.5.5.4. Club List



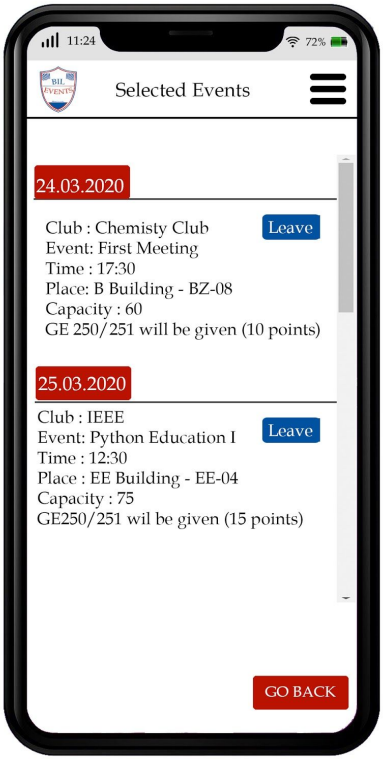




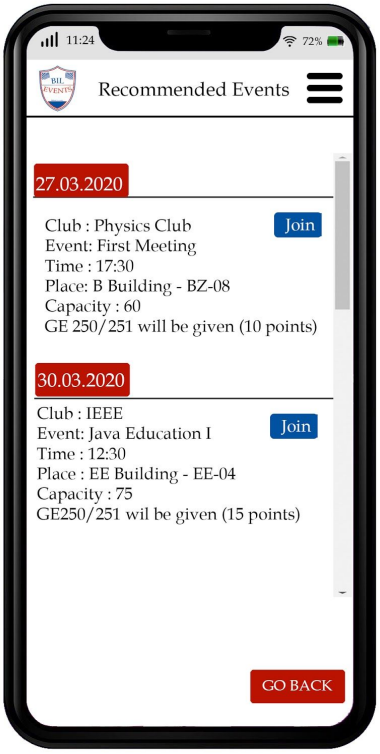
3.5.5.5. Upcoming Events



3.5.5.6. Selected Events



3.5.5.7. Recommended Events



4. Other Analysis Elements

4.1. Consideration of Various Factors

4.1.1 Social Factors

Our app is mostly constrained by social factors. Students in Bilkent University may be concerned about the leak of sensitive data or demonstration such as notification about the events they missed or are not even notified then missed. Students may also be concerned about lack of event information like the current system includes posters and mails but not effective. These concerns are reflected in the design of our system, for instance by making the notification system and recommendation system so that students can involve the events more frequently while being notified until the event. Furthermore, students should not roam all the university and look for events from posters which are not equally distributed. On the other hand, all notification systems will be so easy for them that they will profit from paper and time. Therefore, both students and club leaders should be cared more and both of them collaborate to extract the best possible experience from the ongoing event.

4.1.2. Environmental Factors

As we live in the technology age and there is lots of paper waste, we want to notify people about events while saving paper which is stuck all around the campus regularly. This paper wastage is unnecessary instead of the fact that we offer online notification which is created with the purpose of green world without unnecessary waste

4.1.3. Economic Factors

The performance of the application may be constrained by economic factors. As a result of environmental effects of our application, we aim to descrease paper usage for posters. Therefore, while doing that, we also want to prevent waste of money for posters. Online platform does not require any money from both users and club leaders. So this application will be free for everyone with its maximum performance. In addition to these issues, the application determines how many students can potentially attend the specific event; therefore, smaller events are paid less compared to the bigger events which are paid more. The feature can optimize the number of participants - allocatable resources equilibrium; therefore, it can create the savings on Bilkent University.

4.2. Risks and Alternatives

4.2.1. Failure to Provide Sufficient Accuracy Rate

Considering that as per club leader does not add the event in time and add in the limited time to the event, we have a risk that we will not be able to write algorithms good enough to produce accuracy rates over a huge percentage. Faulty diagnosis leads to wrongly recommended events, therefore delaying needed treatment and in some cases even causing the event is not notified. If this is the case, club leaders will always have a secondary option to add an event in limited time with losing points for their profile but it is not expected from club leaders and also not recommended for not malpracticing of their duty. Low accuracy on recommending the wrong event can emerge in another way as well, for the new users, we will have limited prior information on their preferences; therefore, the recommender system may recommend undesired events to collect the information and update the beliefs about the preferences of the new users. The situation could cause resentment on the new users who don’t have any comprehensive knowledge about how a recommender system works. Thus, we could potentially expect the loss of the number of users.

4.2.2. Scheduling Risks

Implementing an event application like Bil-Events might result in students failing to manage and find the event correctly. We believe organising work packages, a gantt chart about what will do will help us to distribute work properly and set deadlines to follow. However, if a scheduling problem happens, we can manage to finish the prototype which functions very well and is able to handle the essential tasks.

4.2.3. Leak of Private Data

The private data leak can cause other people from out of the university to participate in events. We have to make sure that private data stays private as it is agreed in Terms of Service and Privacy Policy agreement. In order to keep this feature running as promised, we will pay special attention to what parts of data is displayed to whom. Plan of preventing the leak is to store encrypted data on a database by Key Management Service. There cannot be plan B after having leaked the private data so we have to be careful.

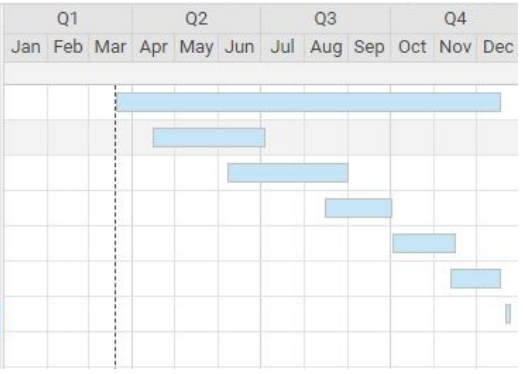
4.2.4 Potential Cheating from Club Members

If we fail to achieve dictating only one account for one student rule, the club members could potentially cheat the system by adding fake accounts or bots to glorify their events. It could potentially cause the unfair competition between cheater club members and other club members. However, we can solve this potential problem by collaborating with Bilkent University.

4.3. Project Plan

Our project plan is represented by Gannt Chart.

	Task Name	Duration	Start	Finish
1	Documentation	273d	20.03.2020	17.12.2020
2	Front End Development	80d	15.04.2020	03.07.2020
3	Back End Development	86d	07.06.2020	31.08.2020
4	Database Implementation	48d	15.08.2020	01.10.2020
5	Recommender System Im	45d	02.10.2020	15.11.2020
6	Testing and Beta Launch	36d	12.11.2020	17.12.2020
7	Launch of the Project	4d	21.12.2020	24.12.2020
8				



4.4. Ensuring Proper Team-Work

In order to have every member involved and engaged in the project, we decided to use the following teamwork tool.

Github

Github will be used for version control and to keep track of the contribution of each team member and changes made to every iteration. It will be the main tool to coordinate work among collaborators.

Discord

Discord will be used for ensuring proper communication between each team members and changes will be shared by communication via voice channel. It will be the main tool to communicate with each other.

Skype

Skype will be used for communicating with the innovation expert and progresses will be shared by communication via voice channel. It will be the main tool to communicate with each other.

4.5. Ethics and Professional Responsibilities

We will collect data from both club leaders and students in order to offer Bil-Events' services. Many of those data contain sensitive information like: student and club leader's ID numbers and . It is crucial for us to be able to save and process those data in total confidentiality in accordance with GDPR (General Data Protection Regulations) and on the grounds of user's agreement to the terms and policies. When detecting a wrong ID number in a certain database, we will not disclose the exact account where ID number has occurred. This will be done in order to prevent users from outsiders. It is very important for us to be able to offer to notify Bilkent University students.

4.6. New Knowledge and Learning Strategies

In order to implement our application we firstly need to learn how to develop applications for the Android Operating System using Android Studio better than before. In order to do that we will use online tutorials and try doing simple examples before we begin working on it. Besides that we have to learn how to implement a Recommendation Engine

for our system. Moreover, encrypted data is also a concern that we have to learn for our users' safety. While implementing our project, we can try and enhance our algorithms to find fastest and the most efficient ways.

5. References

[1] "Java & Android SDK", Predictionio.apache.org, 2020. [Online]. Available: http://predictionio.apache.org/sdk/java/. [Accessed: 24- Feb- 2020].

[2] "Create a Movie Recommendation App with Prediction.io - Setup — SitePoint", SitePoint, 2020. [Online]. Available: https://www.sitepoint.com/create-movie-recommendation-app-prediction-io-setup/. [Accessed: 24- Feb- 2020].

[3] "What Is MongoDB?", MongoDB, 2020. [Online]. Available: https://www.mongodb.com/what-is-mongodb. [Accessed: 24- Feb- 2020].

[4] "Event Server Overview", Predictionio.apache.org, 2020. [Online]. Available: https://predictionio.apache.org/datacollection/. [Accessed: 24- Feb- 2020].

[5] 2020. [Online]. Available: https://predictionio.apache.org/resources/faq/. [Accessed: 24- Feb- 2020].

[6] "Database Failure - an overview | ScienceDirect Topics", Sciencedirect.com, 2020. [Online]. Available: https://www.sciencedirect.com/topics/computer-science/database-failure. [Accessed: 24- Feb- 2020].

[7] "Encryption at Rest — MongoDB Manual", Docs.mongodb.com, 2020. [Online]. Available: https://docs.mongodb.com/manual/core/security-encryption-at-rest/. [Accessed: 24- Feb- 2020].